

Content:

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Data can be defined as:

facts, such as numbers, words, measurements, observations and statistics collected together for reference or analysis

Data analysis is the process of cleaning, transforming, and modelling data to identify patterns, trends and themes. Data analysis is performed to identify connections, and relationships in the data. Stated more simply, the purpose of data analysis is to extract useful information from the data.

The analysis methods differ for qualitative and quantitative research. The data that stems from qualitative research is usually in the format of voice and video recordings, transcripts, observational notes and other high *definition* content. Quantitative data is converted into numerical format that can be analysed using mathematical and statistics models.

Structured and unstructured data

Data that exists in neat and orderly format such as, a spreadsheet is structured data. Unstructured data is relatively disorganized and the researcher has to *clean* the data beforehand.

“Estimates say that just 20% of data is structured, while unstructured data accounts for 80-90% of data regularly generated” (Mesevage, 2020)

Structured data is quantitative or in numerical format, neatly organized, and is easier to analyze using data analytics software. Unstructured data is *messy*, resulting from video recordings, tweets, posts on a website or blog and text narratives. This type of data needs to be cleaned, sorted and categorized before analysis.

Analysing qualitative data

Qualitative data stems from the responses of participants during interviews, focus groups or open-ended questionnaires. The focus is in the interpretation of ideas, opinions and perspectives. It is best suited for the early exploratory stages in the research. Qualitative data differs from quantitative data which is represented in numerical format. Qualitative data is harder to analyse when compared to quantitative data due to the large amount of *transcribed* data. The researcher would have collected recordings of voice or video and field notes from observations.

The nature of qualitative data analysis

Qualitative data analysis is mainly used for exploratory type research. The researcher is trying to understand the object of interest through people's perceptions, experiences and their *as-lived-reality*. Hence, the collected data is often transcribed into narratives from which different analysis methodologies can be applied. A researcher may not always achieve an *absolute* meaning from qualitative data. This is due to the data being derived from opinions and perspectives.

Qualitative data can be sorted and arranged into *categories* which can then be assigned numerical values. Thereafter, statistical analysis can be performed on the data (described in the section later on). This approach is called mixed-methods and brings rigour to the analysis of qualitative data (which can sometimes become storytelling with no scientific backing).

Qualitative data analysis in IT research is mostly used during the early stages of analysis. It allows for the dominant themes and/or categories to be identified. Thereafter statistical or mathematical methods are applied.

There are several methods for qualitative data analysis. Some of the more popular are content analysis and grounded theory.

Content analysis

Content analysis is used to identify recurrent data. It allows the researcher to discover which are the most frequent occurrences in the collected data. For example, in sentiment analysis the researcher may be interested in particular phrases or words. Content analysis can help determine this.

The researcher would go through the transcribed data and group it into codes, summarise into categories and themes, and then tabulate. This then allows for the easy identification of recurrent and most recurrent themes.

Grounded theory

Grounded theory is used to generate *theory* from collected qualitative data. There are iterative process steps that drive Grounded theory. (See: O' Hagan and O' Connor, 2015: https://www.researchgate.net/figure/Grounded-Theory-Data-Analysis-Steps_fig1_282943329). This iterative process usually involves:

- i. Transcribe the data. This is done by converting the collected data into text. Recordings, observations, images and so on are converted into a text format. This text can then be captured into a data analysis management software tool.
- ii. Arrange the data. The textual data is grouped. The grouping should be derived from the research objectives or questions.
- iii. Data coding. As the researcher examines the data more closely, certain patterns, categories and themes begin to emerge. Through this iterative process, the collected data is gradually *coded* or assigned into categories. Eventually patterns and themes emerge from the data. Recall that qualitative research is exploratory hence, the researcher cannot have preconceived notions and must remain unbiased no matter what is emerging from the data.
- iv. Validate data. The researcher has to proof that the results that emerged from the data are accurate and scientifically repeatable.

Coding in Grounded Theory

Coding is a key step in Grounded Theory. It is a process for transforming textual data transcribed from data collection into codes and then categories until the eventual theory is generated for the study. There are three steps described below. A useful url is:

<https://delvetool.com/blog/openaxialselective>

Open coding: is the first step in coding. The transcripts of data are divided into parts or groups each labelled with a unique code. Open coding because the researcher is required to “open up” to new possibilities and not be biased by preconceived notions or expectations. As this process continues, more data is added to each label.

Axial coding: The second step is when the researcher begins to draw relationships between the codes and begin to organize them into categories. A number of categories will be created which form the “axes” around which supporting codes will be grouped.

Selective coding: last step where the researcher connects all categories into one core category. This core category forms the main theory of the research.

Qualitative data analysis software

Qualitative data analysis software is specialised tools that help the researcher to manage the data analysis process. These specialized tools provide functionality for storing, converting and transcribing, conversion to different formats, content analysis, data mapping, data linking, data coding, iterative abstraction, grounded theory methodology and reporting. This category of software is a must have for qualitative research to save resources and improve on the accuracy.

Some popular examples are: Nvivo and Atlas-Ti

Analysing quantitative data

Quantitative data analysis is performed on data that is numeric or on data that can be easily converted to numeric format. For example, participant responses to questions such as “state your gender” or “what is your age group” will result in data that can be converted to numbered categories. Quantitative data analysis is used to measure the differences and relationships between categories or to test hypothesis.

Statistics is used to perform data analysis of quantitative data. Is concerned with describing, interpreting and analyzing numerical data. There are two main types of statistics, *descriptive* and *inferential*. Descriptive statistics aims to describe the details of the sample (not the entire population). In most cases of quantitative research, descriptive statistics is done *first* and then inferential statistics. Inferential statistics involves more advanced techniques which allow the research to proof that the results are genuine i.e. not fake or doctored and that the results can be generalized to the population.

Descriptive statistics

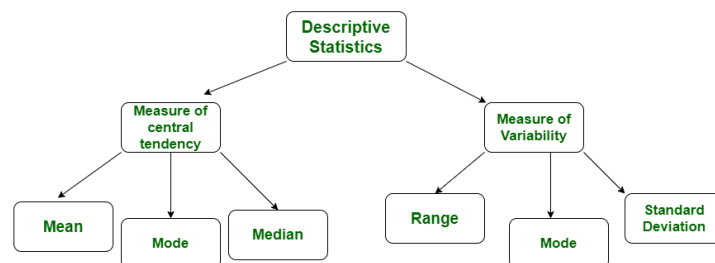
Descriptive statistics is used to organize, summarize and describe data. For example, consider a class list with final marks. The first step would be to organize the marks into ascending/descending order. Then the average mark could be determined to see how well the class performed. The number of failures and distinctions. This summary tells informs us about the performance of the class and interventions can be put into place by the lecturer.

The researcher also performs descriptive statistics to identify any errors in the data and to determine the level of *skewness* in the data. The skewness is used to determine the type of inferential statistic technique that can be used for data analysis. If this part is ignored it could lead to flawed results.

Basic descriptive statistics are grouped into *measure of central tendency* and *measure of variability*. Measure of central tendency allows for the identification of the center point or average of a dataset. For this purpose, *mean, median and mode* are calculated. The measure of variability is a measure of dispersion in the dataset or stated differently, how far away the data points are from the center point. For this, *range, mode and standard deviation* is calculated.

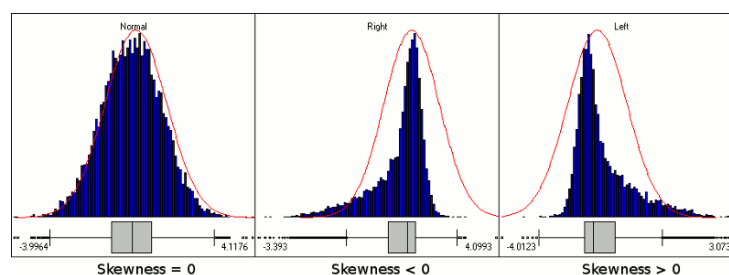
- Mean – average of a range of numbers
- Median – middle point of a range of numbers (if those numbers sorted)
- Mode – the number that appears most often
- Range – the difference between the largest and the smallest number
- Standard deviation and variance – these indicate how dispersed a range of numbers are. In other words, how close (or far) all the numbers are to (or from) the average.

The diagram below is from Geeksforgeeks (2020):



Skewness indicates how symmetrical a range of numbers is. In other words, do they tend cluster into a smooth bell curve shape in the center point (this is called a “normal distribution”) or do they skew to the left or right. Skewness shows the distribution of data from the mean.

The diagram below from Delve.com:



Linear models work on the assumption that the distribution of the independent variable and the target variable are similar. Therefore, knowing about the skewness of data helps in creating better linear models (<https://www.analyticsvidhya.com/>)

Below is an example of basic descriptive statistics from Jansen and Warren (2020)

Data set		Descriptive statistics		
Person	Weight (kg)	Statistic	Result	Formula
1	70	Mean (average)	72.4	=AVERAGE
2	55	Median	74	=MEDIAN
3	90	Standard Deviation	10.637	=STDEV.S
4	75	Variance	113.15556	=VAR.S
5	66	Skewness	-0.223701	=SKEW
6	73			
7	80			
8	58			
9	76			
10	81			

The formula in the image above are from Excel.

Inferential statistics

Inferential statistics allow for the researcher to make inferences about the population but based on the sample. This is a prediction of what the researcher might find in the population. The predictions could include *differences* between groups and *relationships* between variables. A difference could be *the difference between assessment marks of IT students between those with laptops and those without*. A relationship could be *the assessment mark and number of hours a student spends doing tutorials on their laptop*.

One type of Inferential statistics is *hypothesis testing*. Based on the examples above, a hypothesis might be stated as

H1_o: There is no difference in marks between those students with laptops and those without

H1_a: There is a difference in marks between those students with laptops and those without

H2_o: students who spend more time doing tutorials on their laptops achieve higher marks

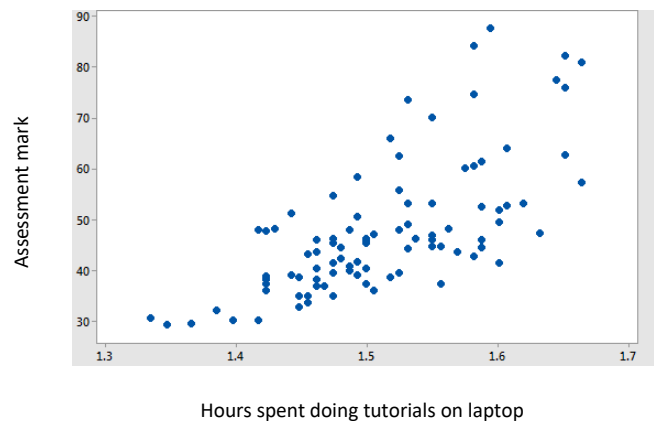
H2_a: students who spend more time doing tutorials on their laptops do not achieve higher marks

Statistics inform the researcher which hypothesis is true and which are false.

Some basic inferential statistics include:

- T-test – the sample is split into *two* groups. T-test compares the averages of two groups of data to determine the level (magnitude) of difference. In other words, are the group averages significantly different in mean, standard deviations and skewness.
- ANOVA – similar to a T-test, but it allows the researcher to analyse more than two groups.
- Correlation – assesses the relationship between two variables. In other words, if one variable goes up does the other variable also go up, down, or stay the same.
- Regression – this is similar to correlation but, it goes a step further to understand cause and effect between variables, not just whether they move together. In other words, does the one variable actually cause the other one to move, or do they just happen to move together naturally due to another cause.

The graph below maps the assessment mark for each student, that is each student is represented as a blue dot. This is called a scatter graph.



The positioning of each plot (each student shown as a blue dot) indicate some relationship between the variables (marks and hours spent). According to the *splatter*, the majority of students spend just less than 1.5 hours per day doing tutorials on their laptop and their mark is between 30% and 50%. Those that spend more than 1.6 hours per day have marks in between 40% and 90%.

A diagonal line can be imagined from bottom-left to top-right of the graph. – *Regression line*.

There is a wide range of inferential statistical techniques. The choice is determined by the *type* of data and the *research question or hypothesis*. Different inferential statistical techniques work with different type of data that is, categorical data (ordinal and nominal) and numerical data (interval and ratio).

Numerical data

Numerical data is numbers based and the variables naturally occur in numerical format for example, student *study period* or *final mark*.

Categorical data

Categorical data must belong to a category first before being assigned a numerical format. For example, final mark between 0 and 50 is category 1, between 49 and 60 is category 2, and between 59 and 101 is category 3. In other words, the final marks of all students have been placed into categories and assigned a numerical value of either 1, 2 or 3.

Interval and ratio data

Numerical data can be either **interval** or **ratio**. **Interval** data is naturally quantitative that is, it naturally exists as numbers (for example final mark). Interval data has a ranking and the options are equally spaced apart. Also, the zero point is arbitrary that is, a zero is a value and not a *null* or *missing value*. Consider a temperature reading: -2; -1; 0; 1; 2. Zero temperature is an actual data value and the options are equally spaced apart. **Ratio** data is similar to interval data with the exception that a zero value means *no* or *null* instances of that option. Stated differently, zero means that there is nothing of that variable in the sample. Ratio data is evenly spaced and it is ranked.

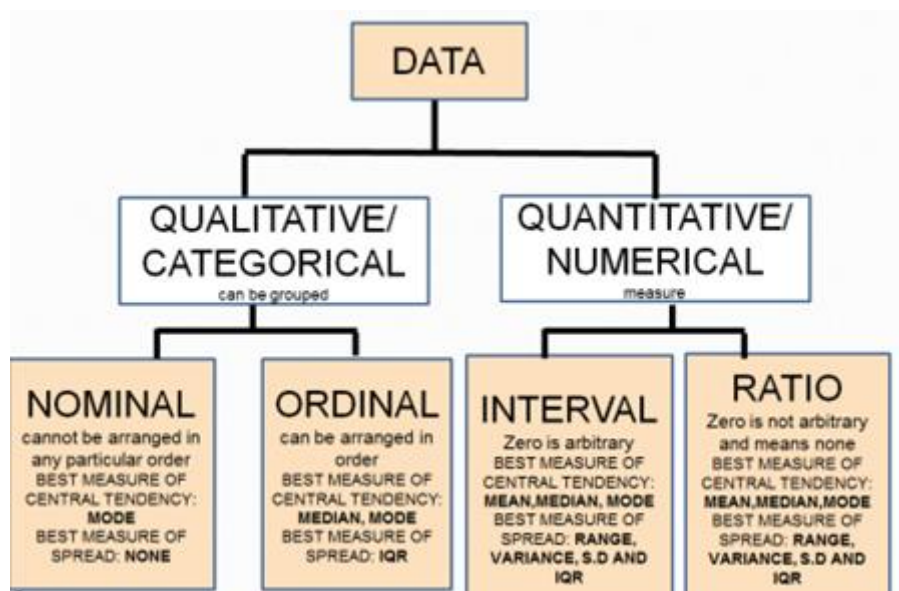
Ordinal and nominal data

Categorical data can be either **ordinal** or **nominal**. **Nominal** data does not have a ranking or natural order. All data are equal but different. For example, data about gender, race and marital status do not have a ranking and one response cannot be better than the other. **Ordinal** data does have a ranking

and a difference. For example, level of agreement can be “disagree, neutral or agree”. The data collected for *level of agreement* is a category but, the difference between the options (disagree, agree and neutral) cannot be numerically measured. They can be ordered and logically ranked.

Provides:	Nominal	Ordinal	Interval	Ratio
The “order” of values is known		✓	✓	✓
“Counts,” aka “Frequency of Distribution”	✓	✓	✓	✓
Mode	✓	✓	✓	✓
Median		✓	✓	✓
Mean			✓	✓
Can quantify the difference between each value			✓	✓
Can add or subtract values			✓	✓
Can multiple and divide values				✓
Has “true zero”				✓

Characteristics of data types. Source: Ragunath (2019).



Data types. Source: Ragunath (2019)

Software tools for quantitative data analysis

Popular tools for quantitative data analysis include SPSS, Stata, Matlab and R. These software packages provide tools for the organization, analysis, interpretation, and presentation of data.

Tutorial

1. What is the difference between primary and secondary data?
2. Use MS Excel to calculate descriptive statistics of a set of 10 test marks.
3. Create a simple question that uses radio buttons for options that have nominal responses. Use an online survey tool.
4. Create a simple question that uses radio buttons for options that have ordinal responses.
5. How does the zero value differ in interval and ratio data?
6. If the research aims to understand the characteristics of a sample, which type of statistical testing is better suited? If aimed to understand the entire population?

References

Jansen, D. and Warren, K. 2020. Quantitative data analysis 101. Available: <https://gradcoach.com/quantitative-data-analysis-methods/>. Accessed: 19 February 2021.

Ragunath, D. 2019. Data levels of measurement. Available: <https://medium.com/@rndayala/data-levels-of-measurement-4af33d9ab51a>. Accessed: 23 January 2021.

Useful urls

<https://delvetool.com/blog/openaxialselective>

<https://gradcoach.com/nominal-ordinal-interval-ratio/>

<https://medium.com/@rndayala/data-levels-of-measurement-4af33d9ab51a>

<https://monkeylearn.com/blog/structured-data-vs-unstructured-data/>

<https://www.questionpro.com/blog/qualitative-data/>

<https://www.analyticsvidhya.com/>